Review of Anthropic's "Risks from Automated R&D" Section: A Detection Engineering Analysis of an AI Safety Policy Document

Abstract

This working paper examines a document published by METR on November 5, 2026, which reviews the "Risks from automated R&D" section of Anthropic's public August 2026 Risk Report. The review was conducted as part of a high-access assessment in which Anthropic shared a private version of the report alongside extensive non-public data. METR agrees with Anthropic that catastrophic risks from models available by July 15th were low, but argues it is plausible that an early warning threshold, the 2x Acceleration Threshold, had already been crossed. The document debates benchmark methodology, counterfactual acceleration estimation, and the interpretation of capability metrics such as AECI and unnamed leading indicators. From a detection engineering standpoint, this document contains no security content. It describes no attacker behaviour, no intrusion technique, no observable adversary action, and no security telemetry. The only rule produced is a null rule confirming that no applicable detection source exists. This paper explains why the document was routed to a detection queue erroneously, what a defender cannot see from this class of material, and what limitations apply to any analysis built solely from the provided source text.

What happened

METR, an organisation described in the source as Model Evaluation and Threat Research, published a blog post on November 5, 2026, titled "Review of the 'Risks from automated R&D' section in Anthropic's Risk Report (August 2026)." The post reviews one section of a risk report that Anthropic had published publicly in August 2026. The review was conducted under a high-access arrangement in which Anthropic shared a private, unredacted version of the Risk Report with METR, along with what the source describes as "extensive other non-public data."

The engagement began on August 12, 2026, when Anthropic started sharing confidential information with METR. On August 14, Anthropic shared the unredacted August AI R&D Risk Report. METR delivered a first draft of its review on September 18. Between September 22 and October 28, METR iterated on the review. Anthropic provided final redaction guidance on October 28 and approved the redacted report on October 29. The post was published on November 5.

The source states that "the exact scope of the review was never formalized, but mutually negotiated over the course of the engagement." Anthropic held rights to redact non-public information from the published post. METR notes that "all redactions that were important to our conclusions are described in the report below."

Anthropic's Responsible Scaling Protocol, version 3.4, sets two capability thresholds for "Automated R&D in key domains." The first, which METR names the Full Substitution Threshold for clarity, is triggered if Anthropic develops a model that can "fully substitute for our entire set of Research Scientists and Research Engineers, at competitive costs (i.e., within a factor of 5)." The second, which METR names the 2x Acceleration Threshold, is triggered by "dramatic acceleration" of the pace of AI progress for reasons that likely relate to the automation of AI R&D. METR notes that these threshold names do not appear in the RSP itself and were assigned for clarity in the review.

The Risk Report argues that the Full Substitution Threshold had not been crossed because models were not currently substituting for humans. Anthropic states in the report: "we believe it would

be clear to us if models were capable of substituting, even at high cost, for the technical work of our entire research staff, and the fact that we do not perform such substitution even when strongly incentivized to is evidence against our models having this level of capability." METR finds this argument "broadly compelling," noting that the threshold is a high bar and that Anthropic would likely have noticed greater progress on automating research if models as of July 15th were capable of this. However, METR's confidence is reduced by "the lack of high-confidence unsaturated benchmarks demonstrating capability limitations that seem critical for automating AI R&D."

The 2x Acceleration Threshold is where METR's disagreement with Anthropic is most pronounced. The Risk Report presents three primary lines of evidence that this threshold had not been crossed. The first concerns limited acceleration in leading indicators. The public report states: "Our leading indicators point to a picture of meaningful acceleration starting in early-to-mid 2025, though by less than a factor of 2." The private report contains a redacted figure that METR calculates implies an acceleration rate of 1.97. METR states that "the 1.97 figure does not appear in the private report; this is our calculation that combines the report's estimate of the current rate of progress above with an estimate of the historical rate of progress elsewhere in the private report." METR argues that a reader of the public report would interpret "less than a factor of 2" to mean a figure meaningfully below 2, and that 1.97 would be better described as "around 2." METR's own analysis, which was redacted from the published report, finds "a range of reasonable estimates, between 1.3 and 2.8."

The second line of evidence concerns attribution of acceleration to AI. The public report states: "We are fairly confident in attributing the acceleration in 2025 to factors other than our use of AI models, though we also believe that our AI models have been a key factor in the faster trends continuing through the coverage date." The private report adds: "Our central estimates of the speedup of AI R&D progress relative to an unassisted counterfactual are slightly less than a factor of 2 for [redacted]. … we would guess this is more likely to be an overestimate than an underestimate; nonetheless, we have enough uncertainty that it seems plausible we do exceed this factor of 2 overall." METR notes that the private report describes it as "plausible" that overall counterfactual acceleration attributable to AI exceeds a factor of 2 by July 15th, while the public report implies reasonably high confidence that only a smaller fraction is attributable to AI.

The third line of evidence concerns AECI, a capability metric referenced in the report. The public report states: "Claude Mythos 5 lies above the historical score trendline by a similar amount to Mythos Preview, which is weak evidence against a dramatic increase in overall model capability trends over time. Based on limited data, Model 2 (not shown on this figure) appears to be around 1.5 points higher on AECI than Claude Mythos 5, with large error bars (a smaller increase than the one from Claude Mythos Preview to Claude Mythos 5)." The private report adds qualifications: "Given that not all of our models are trained end-to-end to maximize intelligence per FLOP (indeed, none of them are exclusively optimizing for this goal), there is a question of how many 'effective points' we should consider this [AECI] figure to have, and whether the apparent strong linear trend is as reliable a pattern as it appears." The private report also notes that Anthropic expects higher scores from later snapshots of Model 2's base model. METR states that "further snapshots of Model 2 did improve on AECI by about the same amount as Mythos 5 did over Mythos Preview."

METR also discusses the distinction between a one-time jump in AECI and a continued acceleration. METR states that "if we reuse the data that was used in the Risk Report, the estimated acceleration is around 1.5x." After the Risk Report's publication, Anthropic added data from more benchmarks, which "estimates an acceleration of 1.9x using the same set of models." METR argues that "a continued acceleration in AECI would be more consistent with the continued acceleration observed in the leading indicators."

Despite these disagreements, METR agrees with Anthropic's overall risk designation of "low" for July 15th models. METR states: "While we think it is plausible, given the above evidence, that models available by July 15th were providing 2x counterfactual acceleration of AI R&D progress, significantly greater acceleration would likely be necessary to enable the kinds of catastrophic risks described in Anthropic's scenarios of concern above." METR also notes that it does "not believe models at the time could fully automate AI R&D or R&D in other key domains."

The threat model described in Anthropic's Risk Report includes illustrative scenarios involving domains such as "robotics to energy to cyberwarfare to AI R&D itself." The report states that "if AI models develop dangerous goals while heavily automating R&D in key domains, they may cause unbounded harm, up to and including humanity losing control over civilization entirely, by leveraging novel technology and their access to it." The word "cyberwarfare" appears only as an example domain within this threat model, not as a description of any specific attack, technique, or observed adversary behaviour.

Model names appearing in the source text, including Claude Mythos 5, Mythos Preview, and Model 2, are AI model identifiers used for capability benchmarking. They are not threat-actor tooling, malware names, or software with security detection relevance.

The authors of the METR review are listed as Tom Cunningham, Nate Rush, Parker Whitfill, and Hjalmar Wijk. The post references a companion "holistic assessment of AI R&D acceleration at Anthropic" conducted by the same METR employees.

Detection opportunities

There are no detection opportunities in this document. The single rule file named in the rules for this analysis is `01-no-applicable-detection-source-contains-no-securit.yml`, which encodes the conclusion that no applicable detection source exists because the source text contains no security-relevant content.

A detection engineer reviewing this document finds no attacker behaviour to model. There are no technique descriptions that map to MITRE ATT&CK techniques or any comparable framework. There are no observables that map to any security log source, whether endpoint, network, identity, cloud, or application telemetry. There are no indicators of compromise, no file paths, no registry keys, no process names, no command-line arguments, no network connections, no user accounts, no authentication events, and no system calls.

The document discusses "cyberwarfare" once, but only as an illustrative domain within Anthropic's threat model. The full quotation is: "most or all of the work needed to advance research and development in key domains, from robotics to energy to cyberwarfare to AI R&D itself, may become automatable." This is a hypothetical scenario about future AI capabilities, not a description of any cyber operation, intrusion, or observable event. A detection rule targeting the word "cyberwarfare" in this context would produce no useful alert, because no attack is described.

The model names in the text, Claude Mythos 5, Mythos Preview, and Model 2, are identifiers for AI models undergoing capability evaluation. They are not malware families, tool names, or attacker infrastructure. A detection rule matching these strings would be matching internal AI model nomenclature, not adversary tooling.

The metric AECI is a capability benchmark, not a security metric. The leading indicators discussed in the report are measures of AI research progress velocity, not security observables. The coun-

terfactual acceleration estimates compare AI-assisted research speed to a hypothetical unassisted baseline, which is a methodological construct with no telemetry representation.

The rule file `01-no-applicable-detection-source-contains-no-securit.yml` therefore serves as a formal record that this document was reviewed for detection content and none was found. It is the appropriate output for a document that has been routed to a detection engineering queue but contains no detection-relevant material. The rule does not match any log source, does not define any detection logic, and does not produce any alert. Its purpose is to document the absence of applicability, ensuring that the document is not repeatedly re-queued for detection analysis.

What cannot be detected

This document illustrates a category of material that is structurally invisible to security detection engineering. The subject matter is internal AI capability benchmarking, counterfactual acceleration estimation, and Responsible Scaling Protocol threshold interpretation. These are analytical and policy activities that produce no network artefacts, no endpoint observables, no identity events, and no log entries in any security-relevant source.

From the vantage of a security information and event management system, a next-generation firewall, an endpoint detection and response platform, a cloud security posture management tool, or an identity provider, the activities described in this document are entirely opaque. When Anthropic researchers evaluate whether a model can substitute for research staff, the evaluation process may involve computational workloads, API calls, and data processing, but the source text describes none of these operational details. It describes only the conclusions drawn from the evaluation and the methodological disagreements between METR and Anthropic about how to interpret those conclusions.

The counterfactual acceleration estimates are particularly invisible. A counterfactual is a hypothetical construct, a comparison between observed reality and an imagined baseline without AI assistance. There is no log entry for a counterfactual. There is no network flow for a hypothetical. There is no process execution for a methodological judgment call. The entire debate about whether acceleration is 1.5x, 1.9x, 1.97x, or somewhere between 1.3 and 2.8x is a debate about analytical methodology and statistical estimation, not about events that occur on computing infrastructure.

The distinction between a one-time jump in AECI and a continued acceleration is similarly undetectable. This is an interpretive question about trend lines in benchmark data. The benchmark data itself, AECI scores for specific model snapshots, is a product of internal evaluation pipelines, but the source text provides no details about those pipelines, their infrastructure, their logging, or their observable characteristics. Even if a defender had access to Anthropic's internal systems, the source text does not describe what telemetry those systems produce.

The redaction process is also invisible. Anthropic redacted non-public information from METR's post, including METR's own analysis of leading indicators and a discussion of methodological uncertainty. The source text notes that "Anthropic redacted this discussion" and that certain figures are "[redacted]" in quoted material. Redaction is a document-level editorial process, not a system-level event. There is no detection rule that can recover redacted content from a published blog post, and the source text provides no information about the redacted content beyond what is stated.

The private, unredacted version of the Risk Report is described as containing additional detail, including specific progress rate figures, qualifications about model training pipelines, and expressions

of uncertainty that the public report omits. A defender reading only the public report would not see these details. A defender reading the METR review sees some of the differences between public and private versions, but only as described by METR, and only to the extent that Anthropic permitted disclosure. The gap between what Anthropic knows internally and what a defender can observe from public or semi-public sources is a structural limitation that no detection rule can bridge.

From the vantage of an external observer, which is the position of any detection engineer working from this source text, the entire engagement is invisible. The confidential sharing that began on August 12, the unredacted report shared on August 14, the draft delivered on September 18, the iteration period, the redaction guidance on October 28, and the approval on October 29 are all described after the fact in a published blog post. None of these events generated security telemetry available to an external party. Even an internal party at Anthropic with access to collaboration logs would see document sharing, editing, and approval events, but these are standard business workflow events, not security observables.

The threat model described in the Risk Report, which includes scenarios involving cyberwarfare, surveillance, and loss of human control over civilisation, is a forward-looking risk assessment. It describes concerns about what AI models might do in the future, not actions that have been observed. Detection engineering is fundamentally retrospective and observational: it looks for evidence of things that have happened or are happening on systems that produce logs. A hypothetical scenario about what might occur if AI models develop dangerous goals and leverage novel technology is not something that produces a log entry until and unless it manifests as concrete activity, at which point the detection question would be about that specific activity, not about the threat model that anticipated it.

Limitations

This working paper is a machine draft produced for a human editor. It is based solely on the source text provided, which is a single blog post from METR, and on the checked assessment. No additional telemetry was consulted. No internal Anthropic systems were accessed. No METR systems were accessed. No detection rules were tested against real data, because no detection rules were produced beyond the null rule documenting the absence of applicable detection sources.

The source text is itself a secondary document. It is METR's review of Anthropic's Risk Report, not the Risk Report itself. It contains quotations from both the public and private versions of the Risk Report, but these quotations are selected by METR to support its analysis. The full text of neither version of the Risk Report was available to this analysis. The companion holistic assessment referenced multiple times in the source text was also not available. Any facts not present in the source text or the assessment are necessarily absent from this paper.

The redactions in the source text limit the analysis. METR's own analysis of leading indicators, which produced the range of 1.3 to 2.8, was redacted. The specific progress rate figure in the private report, described as "[redacted]x/yr," is not available. The discussion of methodological uncertainty that METR describes as having been redacted by Anthropic is not available. These redactions mean that the source text is incomplete relative to the full analysis that METR conducted, and this paper is correspondingly incomplete relative to what a full analysis might have considered.

The assessment provided with the source text concludes that this document should be skipped for detection engineering purposes. This paper respects that conclusion while still producing a structured analysis, because the working paper format requires coverage of each heading. The detection oppor-

tunities section therefore documents the absence of opportunities rather than describing any actual detection logic. The rule file `01-no-applicable-detection-source-contains-no-securit.yml` is a null rule that records this conclusion.

No reproduction or validation of METR's calculations was performed. The 1.97 figure that METR derives from the private report's estimates is reported as METR's calculation. The 1.5x and 1.9x acceleration estimates for AECI are reported as stated in the source. The range of 1.3 to 2.8 for leading indicators is reported as METR's finding. None of these numbers were independently verified, and this paper makes no claim about their accuracy beyond reporting what the source states.

The document's publication date of November 5, 2026, and the dates in its timeline, are reported as stated in the source. No independent verification of these dates was performed. The author names, Tom Cunningham, Nate Rush, Parker Whitfill, and Hjalmar Wijk, are reported as listed in the source. No independent verification of authorship was performed.

This paper uses British spelling throughout, as required by the working paper format. It avoids em dashes and en dashes, using commas and full stops instead. It contains no placeholders, no notes to the editor, and no preamble before the H1 title.

The fundamental limitation of this analysis is that the source document is not a security document. It is an AI safety policy review. It was written by AI safety researchers for an audience concerned with AI capability thresholds and risk assessment methodology. It was not written for detection engineers, and it does not contain the kind of information that detection engineering requires: descriptions of adversary behaviour, observable indicators, system events, or telemetry patterns. Any attempt to construct detections from this material would require inventing facts not present in the source, which the rules of this working paper prohibit.

References

https://metr.org/blog/2026-11-05-rd-section-anthropic-risk-report-august-2026-review/